Education, Child Marriage, and Heterogeneous Wage Returns among Women in Nigeria

Table 1: Descriptive Statistics – Wage Earners

Variable	National (N=2,387)	N. Nigeria (N=932)
Continuous: Mean (SD)		
Log annual wage	12.02 (1.42)	11.83 (1.60)
Years of schooling	14.42 (5.81)	13.06 (7.02)
Age	36.69 (11.03)	36.34 (10.85)
Experience	16.01 (12.13)	16.65 (11.91)
Binary (%)		
Child marriage (%)	11.7	23.1
Rural (%)	49.9	56.3
Literate (%)	86.1	78.9
Currently married (%)	61.0	70.7
Religion (%)		
Religion: Islam (%)	76.5	53.1
Religion: Christianity (%)	23.0	46.8
Religion: Other (%)	0.5	0.1
Father education (%)		
Father edu: None (%)	33.6	38.1
Father edu: Primary (%)	43.6	28.2
Father edu: Secondary+ (%)	22.8	33.7
Mother education (%)		
Mother edu: None (%)	45.1	51.2
Mother edu: Primary (%)	40.9	28.2
Mother edu: Secondary+ (%)	14.0	20.6
Sector (%)		
Sector: Urban (%)	50.1	43.7
Sector: Rural (%)	49.9	56.3
N	2,387	932

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied.

Table 2: Sample Construction – National

Step	Description	N	N_dropped
0	Full Whole Country dataset	27,488	0
1	Women only	27,488	0
2	Age 18-64 (working age)	27,488	0
3	Non-missing key variables	26,450	1,038
4	Workers with valid wage data	2,387	24,063

Table 3: National – Quadratic Mincer

	Basic	+ Controls	+ State FE	+ Interaction	Heckman
Years of Schooling	0.116***	0.111***	0.089***	0.082***	0.077***
	(0.006)	(0.006)	(0.007)	(0.008)	(0.003)
Years of Schooling²	0.007***	0.007***	0.008***	0.009***	0.009***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)
Experience	0.042***	0.043***	0.040***	0.040***	0.020***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
Experience²	−0.001***	−0.001***	−0.001***	−0.001***	−0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Child Marriage	−0.314***	−0.263***	−0.078	−0.144	
	(0.081)	(0.081)	(0.084)	(0.091)	
Years × Child Marriage				0.022*	
				(0.011)	
Rural		−0.261***	−0.145***	−0.142***	
		(0.044)	(0.048)	(0.048)	
Inv. Mills Ratio ( )					0.822**
					(0.361)
Currently Married		−0.098**	−0.035	−0.035	
		(0.047)	(0.046)	(0.046)	
Observations	2387	2387	2387	2387	26 432
R^2	0.403	0.412	0.479	0.480	
Adj. R^2	0.402	0.411	0.470	0.470	
Residual Std. Error					1.222
State FE	No	No	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	Yes
Interaction	No	No	No	Yes	No
Selection Correction	No	No	No	No	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 4: National – Linear Specification

	Basic	+ Controls	+ State FE	+ Interaction
Years of Schooling	0.169***	0.163***	0.152***	0.156***
	(0.005)	(0.005)	(0.005)	(0.006)
Experience	0.047***	0.046***	0.044***	0.045***
	(0.002)	(0.002)	(0.002)	(0.002)
Child Marriage	−0.120	−0.087	0.031	0.314**
	(0.083)	(0.083)	(0.087)	(0.154)
Years × Child Marriage				−0.025**
				(0.011)
Rural		−0.270***	−0.181***	−0.182***
		(0.046)	(0.050)	(0.050)
State FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
Interaction	No	No	No	Yes
Currently Married		0.087*	0.140***	0.135***
		(0.046)	(0.045)	(0.046)
Observations	2387	2387	2387	2387
R^2	0.362	0.372	0.433	0.434
Adj. R^2	0.361	0.371	0.423	0.424

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 5: Returns by Education Level – National

	By Level
Primary	0.205
	(0.216)
Secondary	0.309**
	(0.122)
Higher Ed	1.558***
	(0.122)
Experience	0.041***
	(0.002)
Experience Sq.	−0.001***
	(0.000)
Child Marriage	−0.167*
	(0.093)
N	2387
R2	0.369

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Omitted: No education. State FE, rural, marital controls included. Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 6: Returns to Literacy – National

	Literacy
Literate	1.159***
	(0.089)
Experience	0.029***
	(0.002)
Experience Sq.	−0.001***
	(0.000)
Child Marriage	−0.242**
	(0.095)
N	2387
R2	0.323

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: State FE, rural, marital controls included. Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 7: Probit: Pr(Wage Employment) – National

	Pr(Wage Work)
Years of Schooling	0.076***
	(0.003)
Years Sq.	0.009***
	(0.000)
Experience	0.020***
	(0.001)
Experience Sq.	−0.001***
	(0.000)
Child Marriage	−0.123***
	(0.040)
Rural	−0.158***
	(0.029)
Observations	26 450
State FE	Yes
Parent Education	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 8: Average Marginal Effects on Pr(Wage Work) – National

Variable	AME	SE	p
Child Marriage	-0.0153	0.0048	0.001
Experience	0.0025	0.0002	< 0.001
Experience Sq.	-0.0001	0.0000	< 0.001
Rural	-0.0209	0.0040	< 0.001
Years of Schooling	0.0098	0.0004	< 0.001
Years of Schooling Sq.	0.0011	0.0000	< 0.001

Note:
Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01. AME via marginaleffects package.

Table 9: Causal Forest: Average Treatment Effect – National

Term	Estimate	Std. Error	t stat	p value
Education Return (per year)	0.155	0.017	9.346	0

Note: Outcome: Log Annual Wage. Treatment: Years of Schooling. 4,000 trees. Source: NLSS 2018-19.

Table 10: Causal Forest: Calibration Test – National

Term	Estimate	Std. Error	t stat	p value
Mean Forest Prediction	1.045	0.184	5.678	0.00
Differential Forest Prediction	0.668	0.355	1.880	0.03

Note: Outcome: Log Annual Wage. Treatment: Years of Schooling. 4,000 trees. Source: NLSS 2018-19.

Table 11: Causal Forest: Variable Importance – National

Variable	Importance
Experience	0.319
Mother's Education	0.230
Age	0.229
Father's Education	0.126
Child Marriage	0.064
Rural	0.032

Note: Outcome: Log Annual Wage. Treatment: Years of Schooling. 4,000 trees. Source: NLSS 2018-19.

Table 12: Sample Construction – Northern Nigeria

Step	Description	N	N_dropped
0	Full Northern Nigeria dataset	16,611	0
1	Women only	16,611	0
2	Age 18-64 (working age)	16,611	0
3	Non-missing key variables	16,140	471
4	Workers with valid wage data	932	15,208

Table 13: Northern Nigeria – Quadratic Mincer

	Basic	+ Controls	+ State FE	+ Interaction	Heckman
Years of Schooling	0.071***	0.066***	0.053***	0.043***	0.048***
	(0.011)	(0.011)	(0.011)	(0.013)	(0.005)
Years of Schooling²	0.012***	0.012***	0.011***	0.011***	0.009***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Experience	0.042***	0.041***	0.040***	0.039***	0.026***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)
Experience²	−0.001***	−0.001***	−0.001***	−0.001***	−0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Child Marriage	−0.262***	−0.237**	−0.150	−0.236**	
	(0.094)	(0.094)	(0.096)	(0.113)	
Years × Child Marriage				0.017	
				(0.012)	
Rural		−0.253***	−0.130*	−0.123*	
		(0.074)	(0.073)	(0.073)	
Inv. Mills Ratio ( )					0.217
					(0.420)
Currently Married		−0.187**	−0.123	−0.113	
		(0.080)	(0.078)	(0.079)	
Observations	932	932	932	932	16 130
R^2	0.540	0.548	0.605	0.606	
Adj. R^2	0.537	0.544	0.594	0.594	
Residual Std. Error					1.021
State FE	No	No	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	Yes
Interaction	No	No	No	Yes	No
Selection Correction	No	No	No	No	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 14: Northern Nigeria – Linear Specification

	Basic	+ Controls	+ State FE	+ Interaction
Years of Schooling	0.172***	0.166***	0.149***	0.155***
	(0.007)	(0.007)	(0.007)	(0.009)
Experience	0.047***	0.046***	0.045***	0.045***
	(0.004)	(0.004)	(0.003)	(0.003)
Child Marriage	−0.142	−0.112	−0.044	0.148
	(0.102)	(0.101)	(0.103)	(0.170)
Years × Child Marriage				−0.018
				(0.013)
Rural		−0.338***	−0.202**	−0.207***
		(0.080)	(0.079)	(0.079)
State FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
Interaction	No	No	No	Yes
Currently Married		0.041	0.060	0.050
		(0.082)	(0.081)	(0.081)
Observations	932	932	932	932
R^2	0.459	0.470	0.539	0.540
Adj. R^2	0.457	0.467	0.527	0.527

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 15: Returns by Education Level – Northern Nigeria

	By Level
Primary	−0.319
	(0.351)
Secondary	0.215
	(0.150)
Higher Ed	1.912***
	(0.139)
Experience	0.044***
	(0.004)
Experience Sq.	−0.002***
	(0.000)
Child Marriage	−0.190*
	(0.105)
N	932
R2	0.521

* p < 0.1, ** p < 0.05, *** p < 0.01

Note: Omitted: No education. State FE, rural, marital controls included. Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 16: Returns to Literacy – Northern Nigeria

	Literacy
Literate	1.529***
	(0.119)
Experience	0.035***
	(0.004)
Experience Sq.	−0.002***
	(0.000)
Child Marriage	−0.309***
	(0.110)
N	932
R2	0.461

* p < 0.1, ** p < 0.05, *** p < 0.01

Note: State FE, rural, marital controls included. Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 17: Probit: Pr(Wage Employment) – Northern Nigeria

	Pr(Wage Work)
Years of Schooling	0.047***
	(0.005)
Years Sq.	0.009***
	(0.001)
Experience	0.026***
	(0.002)
Experience Sq.	−0.001***
	(0.000)
Child Marriage	−0.094*
	(0.050)
Rural	−0.111**
	(0.044)
Observations	16 140
State FE	Yes
Parent Education	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 18: Average Marginal Effects on Pr(Wage Work) – Northern Nigeria

Variable	AME	SE	p
Child Marriage	-0.0080	0.0042	0.056
Experience	0.0023	0.0002	< 0.001
Experience Sq.	-0.0001	0.0000	< 0.001
Rural	-0.0099	0.0041	0.015
Years of Schooling	0.0041	0.0004	< 0.001
Years of Schooling Sq.	0.0008	0.0001	< 0.001

Note:
Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01. AME via marginaleffects package.

Table 19: Causal Forest: Average Treatment Effect – Northern Nigeria

Term	Estimate	Std. Error	t stat	p value
Education Return (per year)	0.149	0.016	9.389	0

Note: Outcome: Log Annual Wage. Treatment: Years of Schooling. 4,000 trees. Source: NLSS 2018-19.

Table 20: Causal Forest: Calibration Test – Northern Nigeria

Term	Estimate	Std. Error	t stat	p value
Mean Forest Prediction	1.054	0.163	6.460	0.000
Differential Forest Prediction	0.968	0.432	2.241	0.013

Note: Outcome: Log Annual Wage. Treatment: Years of Schooling. 4,000 trees. Source: NLSS 2018-19.

Table 21: Causal Forest: Variable Importance – Northern Nigeria

Variable	Importance
Experience	0.380
Age	0.250
Mother's Education	0.161
Father's Education	0.126
Child Marriage	0.046
Rural	0.037

Note: Outcome: Log Annual Wage. Treatment: Years of Schooling. 4,000 trees. Source: NLSS 2018-19.

Table 22: Key Estimates – National vs. Northern Nigeria

Metric	National	N. Nigeria
Average Return (CF)	15.5%	14.9%
Return Std. Error	0.017	0.016
Child Marriage Penalty (OLS+FE)	-7.8%	-15.0%
Interaction: Years x Child Marriage	0.022	0.017
Interaction p-value	< 0.001	0.459
Child Marriage Var. Importance (CF)	6.4%	4.6%
Heckman lambda	0.822**	0.217 (n.s.)
N (wage earners)	2,387	932
N (selection sample)	26,450	16,140

Note: Source: NLSS 2018-19. CF = Causal Forest. OLS+FE = OLS with state fixed effects. *p<0.1, **p<0.05, ***p<0.01.

Table 23: Descriptive Statistics – Full Working-Age Sample

Variable	National (N=26,450)	N. Nigeria (N=16,140)
Continuous: Mean (SD)		
Log annual wage	12.02 (1.42)	11.83 (1.60)
Years of schooling	8.37 (7.07)	5.90 (6.92)
Age	34.57 (12.26)	33.50 (11.69)
Experience	18.90 (14.26)	19.58 (13.69)
Binary (%)		
Child marriage (%)	31.6	46.2
Rural (%)	70.6	77.9
Literate (%)	59.7	47.5
Currently married (%)	69.3	78.4
Religion (%)		
Religion: Islam (%)	51.6	28.3
Religion: Christianity (%)	47.8	71.2
Religion: Other (%)	0.7	0.5
Father education (%)		
Father edu: None (%)	40.1	40.2
Father edu: Primary (%)	29.1	16.0
Father edu: Secondary+ (%)	30.8	43.8
Mother education (%)		
Mother edu: None (%)	50.6	51.3
Mother edu: Primary (%)	25.3	13.1
Mother edu: Secondary+ (%)	24.1	35.5
Sector (%)		
Sector: Urban (%)	29.4	22.1
Sector: Rural (%)	70.6	77.9
N	26,450	16,140

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates (used in first-stage probit). Survey weights applied.

Figure 1: Heterogeneous Returns to Education – National. Causal forest estimates by child marriage status. Source: NLSS 2018-19.

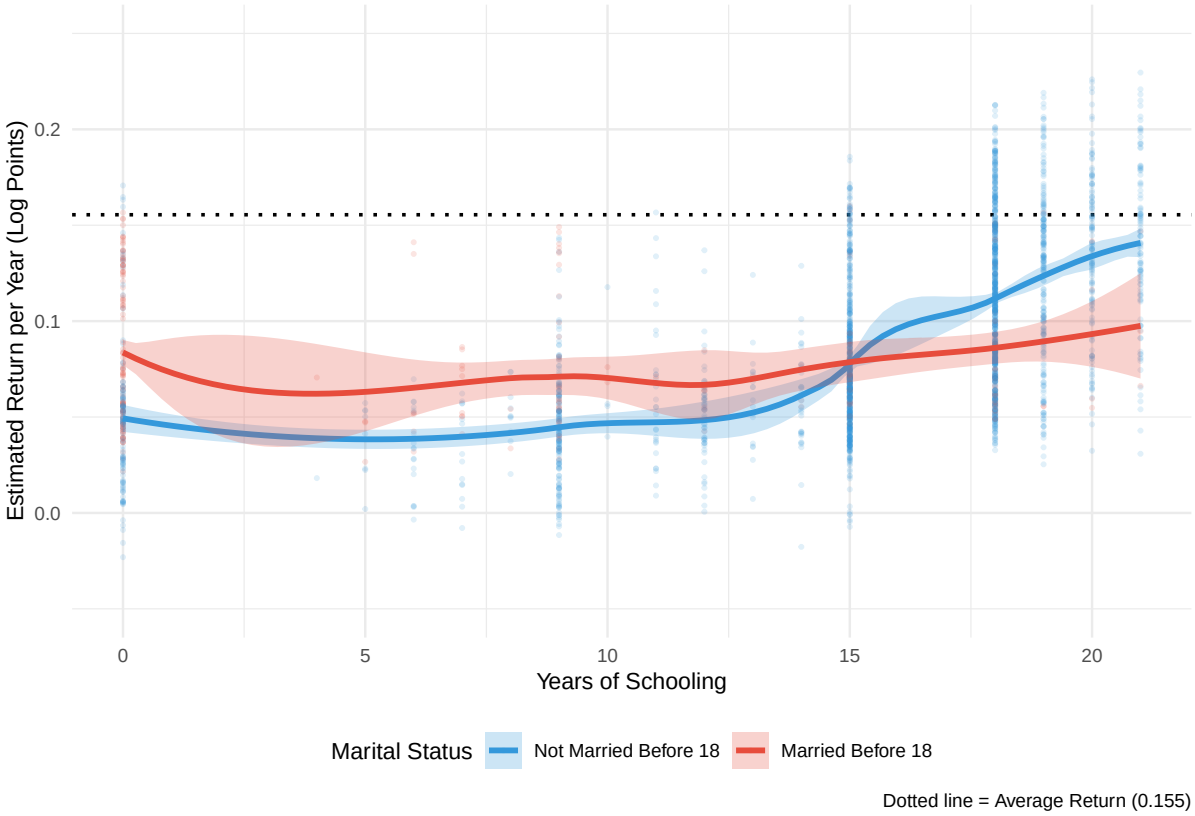

Figure 2: Distribution of Education Returns – National. Density of causal forest returns by child marriage status. Source: NLSS 2018-19.

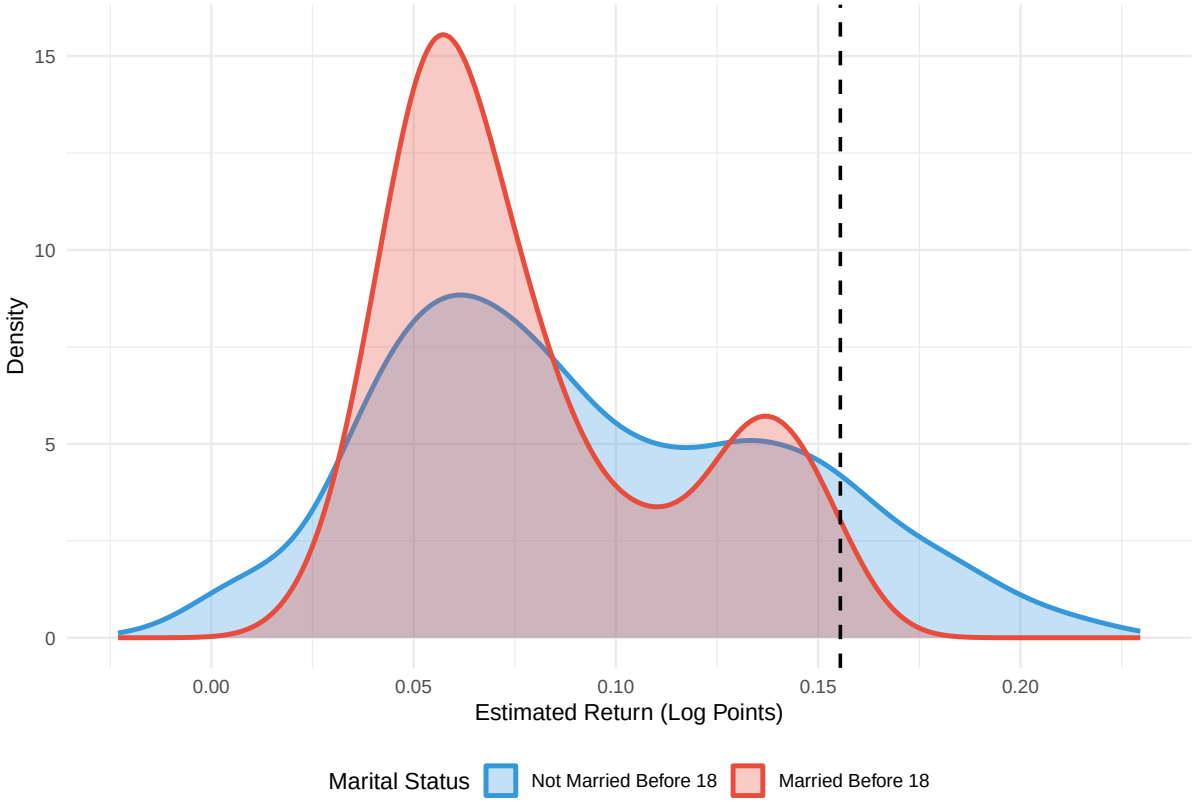

Figure 3: Returns by Level and Marriage Status – National. Average causal forest returns by education level. Source: NLSS 2018-19.

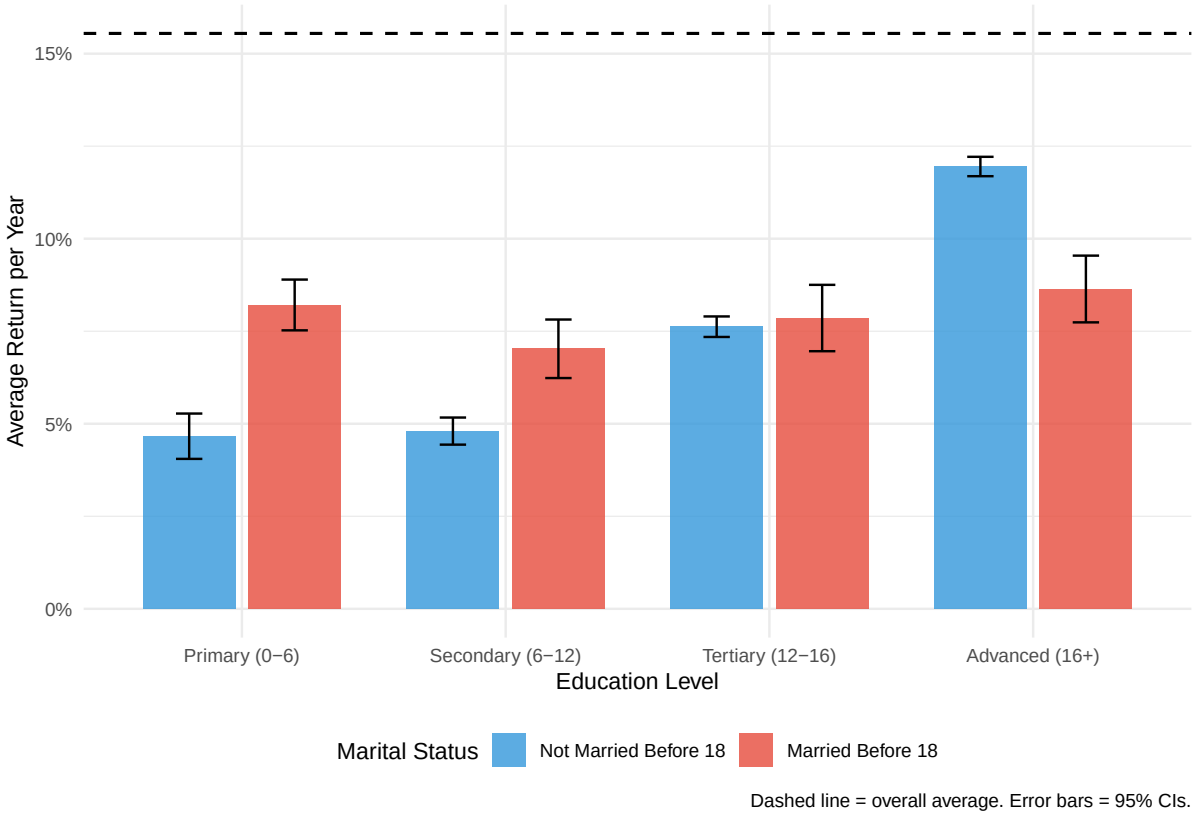

Figure 4: Predicted Returns vs. Wage Outcomes – National. Causal forest predicted returns against wage residuals. Source: NLSS 2018-19.

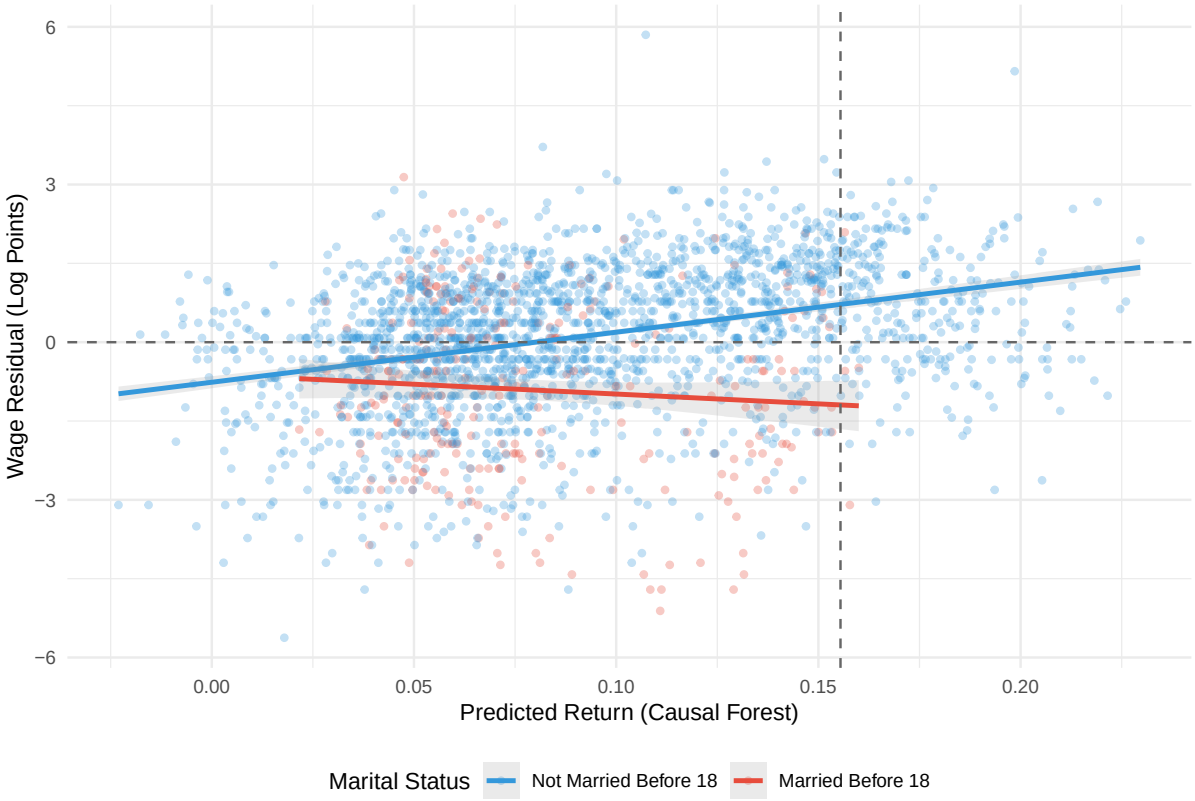

Figure 5: Heterogeneous Returns to Education – Northern Nigeria. Causal forest estimates by child marriage status. Source: NLSS 2018-19.

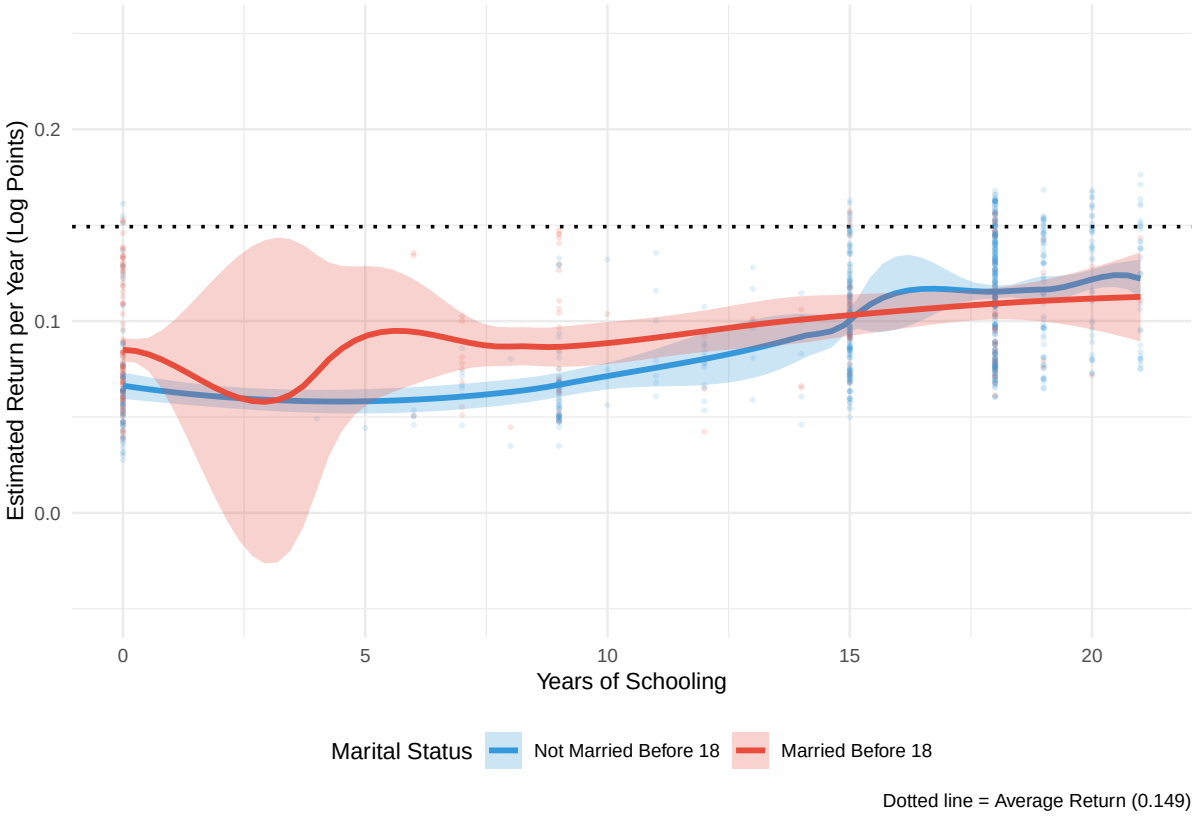

Figure 6: Distribution of Education Returns – Northern Nigeria. Density of causal forest returns by child marriage status. Source: NLSS 2018-19.

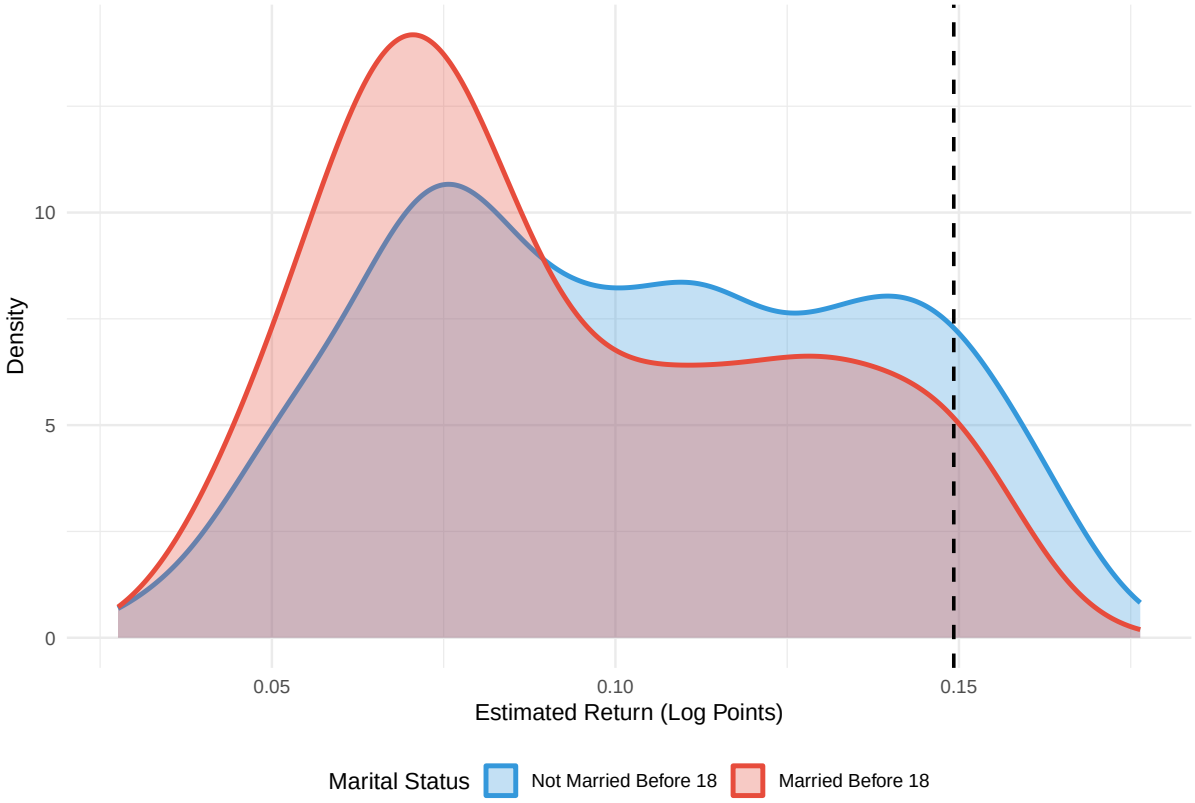

Figure 7: Returns by Level and Marriage Status – Northern Nigeria. Average causal forest returns by education level. Source: NLSS 2018-19.

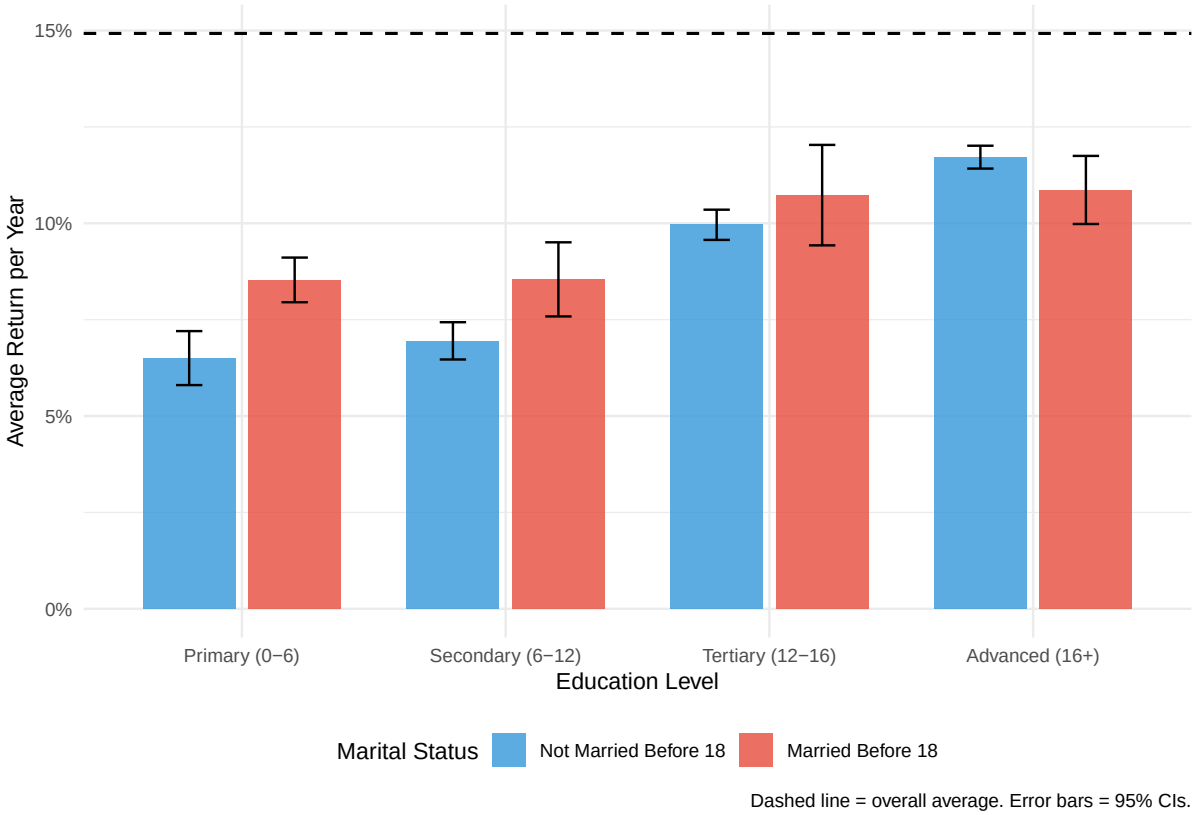

Figure 8: Predicted Returns vs. Wage Outcomes – Northern Nigeria. Causal forest predicted returns against wage residuals. Source: NLSS 2018-19.

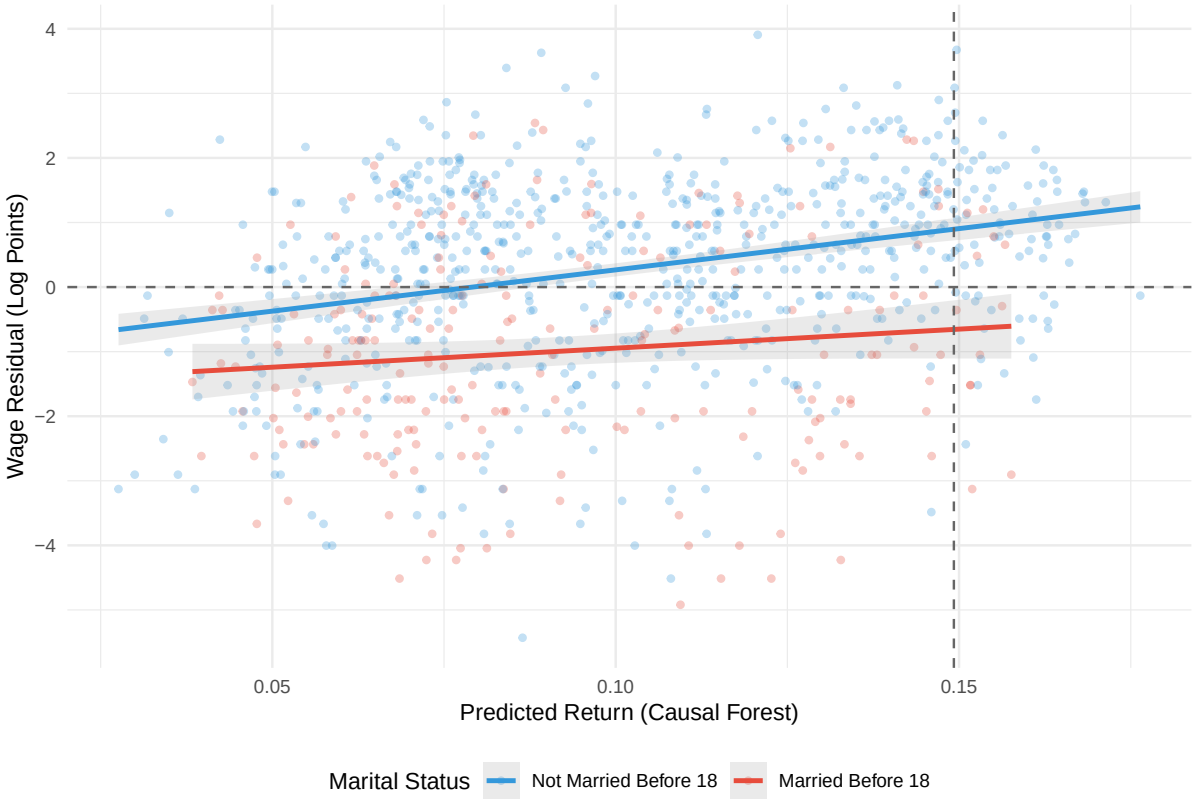

Figure 9: Returns Across Methods – National. Marginal returns from quadratic OLS, linear OLS, and causal forest. Blue = not married before 18; Red = married before 18. Source: NLSS 2018-19.

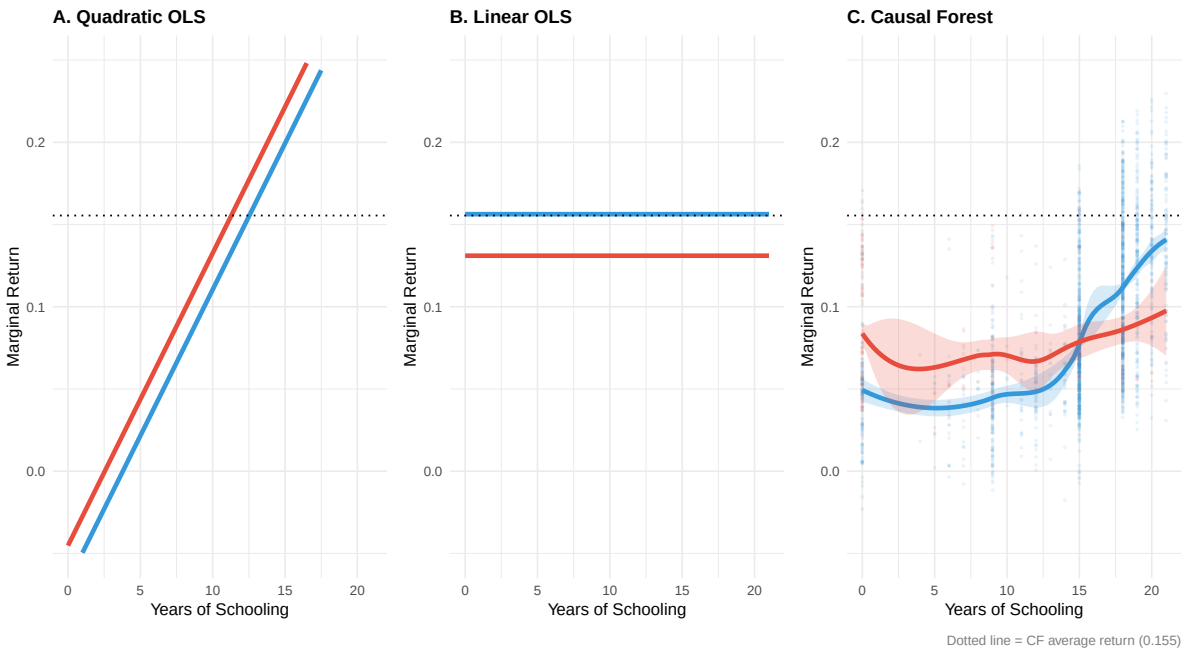
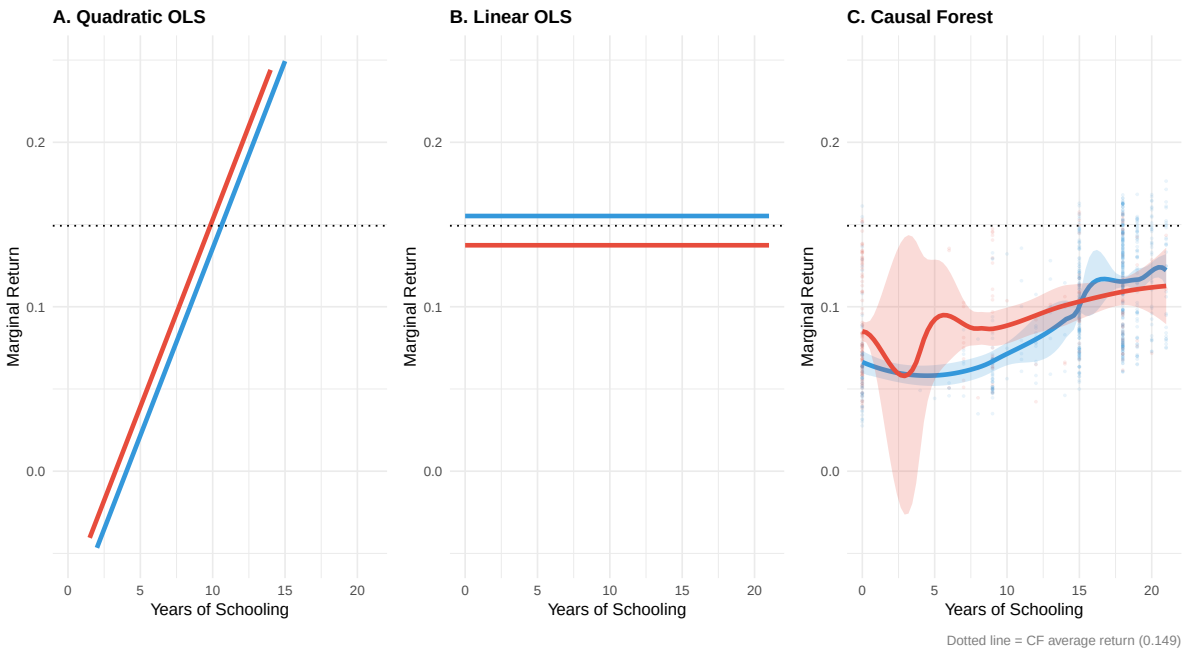
Figure 10: Returns Across Methods – Northern Nigeria. Marginal returns from quadratic OLS, linear OLS, and causal forest. Blue = not married before 18; Red = married before 18. Source: NLSS 2018-19.